

MATH 4790: Quiz 5

April 10, 2026

Name: _____

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted. Credit will not be given to just guessing the correct answer.

Question 1.

A \$1,000 par value bond with annual coupons at a rate of 8% is purchased to yield an annual effective rate of i .

The book value immediately after the first coupon payment is \$900.

The increase in book value from immediately after the first coupon payment to immediately after the third coupon payment is \$21.

Find i .

- (A) 8% (B) 9% (C) 10% (D) 11% (E) 12%

Question 2.

Two bonds are purchased to yield the same annual effective rate j .

The first bond is an n -year bond with face value \$1,000, redemption value \$1,000, and annual coupons at a rate of 6.5%. It is purchased at a premium of X .

The second bond is an n -year bond with face value \$900, redemption value \$1,000, and annual coupons at a rate of 5%. It is purchased at a discount of $3X$.

The total purchase price of the two bonds is \$1,921.13.

Find n .

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Question 3.

A \$1,000 par value bond with redemption value equal to par is purchased for \$927.19 to yield a nominal annual rate of 7.2% compounded monthly.

The bond matures in 8 years and pays level monthly coupons.

Each coupon is deposited into a separate account earning a nominal annual rate of 5% compounded monthly.

What is the accumulated value in the separate account at the time the bond matures?

- (A) \$564.18 (B) \$576.44 (C) \$588.70 (D) \$600.96 (E) \$613.22